

DISPOSITION KINETICS OF CIPROFLOXACIN AFTER DIFFERENT ORAL DOSAGE LEVEL IN RABBITS

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ABSTRACT

Ciprofloxacin a fluoroquinolone antibacterial drug was orally administered to healthy white albino rabbits (n = 12) a single dose of 10, 20, and 30 mg/Kg body weight using a randomized crossover design. Plasma samples were collected at various times over 12 hours following drug administration. The drugs were detected by high performance liquid chromatography (HPLC), and plasma drug concentration versus time curve were subjected to non-compartment and two compartment pharmacokinetic analysis. Descriptive statistics were determined for each doses, and comparison were made among doses groups for selected pharmacokinetic parameters.

Result indicated that increasing dose of ciprofloxacin resulted in increased plasma concentration. Maximum concentration (C_{max}) was 0.43 ± 0.007 , 0.84 ± 0.01 , and 1.46 ± 0.013 $\mu\text{gm/ml}$ for the doses of 10, 20, and 30mg/kg body weight. $AUC_{(0-t)}$, $AUC_{(0-\infty)}$ and MRT were significantly greater at 30mg/kg and showed significant difference with doses ($P < 0.05$), where as other parameters like absorption rate constant, distribution rate constant and elimination rate constant did not show significant difference.

INTRODUCTION

Ciprofloxacin is a second generation Quinolones which possesses activity against a wide microbial spectrum including both gram-negative and some gram-positive bacteria as well as against some Chlamydia, Mycoplasma and many mycobacterium species (Hackbarth *et al.*, 1986; Neu, 1987; Campoli-Richard *et al.*, 1988; Demoleas, 1988; Lefevre *et al.*, 1988). Its action takes place by the inhibition of the A subunit of the bacterial DNA gyrase, an essential enzyme for DNA replication and synthesis. In addition ciprofloxacin is reported to be the main metabolite of enrofloxacin, another antibacterial quinolone (Tyczkowska *et al.*, 1989).

Pharmacokinetic studies on ciprofloxacin have been conducted in humans and some other animal species such as rat monkeys, cats, dogs, and sheeps. Ciprofloxacin was shown to be well absorbed after oral administration, showing a rather high volume of distribution and good tissue penetration.

In animals, ciprofloxacin also exhibits favorable pharmacokinetic properties. Some studies in calves, dogs, cats, rats, chickens and goats indicate that it is rapidly and well distributed through out the body. Its apparent volume of distribution is greater than 1.0 l/kg in all the species studied, suggested substantial tissue penetration. It is eliminated by both metabolism and renal excretion pathways.

The aim of the present study was to determine the influence of the dose on ciprofloxacin pharmacokinetics in rabbits and also to standardize the method for the determination of ciprofloxacin in rabbit plasma, following single dose of the drug: 10, 20, and 30mg/kg body weight.

MATERIALS AND METHODS

Animals: Twelve healthy white albino rabbits of either sex ranging in body weight from 1.0- 1.2 kg were used. All the animals were maintained under similar conditions. The animals were fed with fresh green fodder and black gram in the morning and

evening, while water was provided freely as much they required.

Plan of Study: Pharmacokinetics of ciprofloxacin was studied after administration of a single oral dose of 10, 20 and 30mg/kg body weight in cross-over design on rabbits. A wash out period of 10 days was given for each administration.

Drug Administration: For the determination of disposition kinetics of ciprofloxacin in different dosage rate in rabbits, the drug was administered as a single dose of 10, 20 and 30mg/kg of body weight (Ciprofloxacin tablet of Bayer Pharmaceutical) prepared in solution form. The preparation was administered through the feeding tube orally.

Blood Sampling Procedure: The blood samples were collected through the jugular vein of the rabbits, in heparinized glass centrifuge tubes with the aid of sterilized disposable plastic syringes just before and at 10, 20, 40, 60, 90, 120, 240, 480 and 720 min after the drug administration. The blood samples were centrifuged at 3000 rpm for 10 minutes to separate the plasma for analysis.

Analytical Procedures: The concentration of ciprofloxacin in plasma was determined by the high performance liquid chrom-

atographic procedure as described by Kordick *et al* (1997) with some modified sample preparation technique and few other variables suitably adjusted in the lab. The concentration of ciprofloxacin in unknown plasma sample were determined by comparing the peak areas to that of the standard curve with various concentration of ciprofloxacin (.05-2.0 ug/ml).

Sample Preparation: After separating the plasma from blood sample, equal amount of 5% perchloric acid was added for protein precipitation, vortexed for two minutes then centrifuge at 2000 rpm for 10 minutes. The aliquot was separated for injection into the auto sampler of the chromatographic system (Agilent 1100 series with vacuum degasser, Quaternary pump, Auto sampler 10v, Thermo stated column compartment, Variable wave length detector) and a column (Zorbex Eclipse x DB C 18 Analytical 4.6 x 150mm- 5 μ).

PHARMACOKINETIC ANALYSIS

Pharmacokinetics analysis was performed by non compartmental approach along the compartmental approach as well; using best fitted two compartmental models, using the soft wear, Pharmacolysis® (Ahmed, 2006) provided by Pharma Professional Services.

Table 1
Time vs Plasma level [Mean (12) $\pm$ SEM] of ciprofloxacin following oral administration of single dose i.e. 10, 20 and 30 mg/Kg body weight

Time (hr)	Plasma Levels (μ g/ml)					
	10 mg/kg		20 mg/kg		30 mg/kg	
	Average	SEM	Average	SEM	Average	SEM
0.17	0.31	0.0070	0.68	0.02	0.85	0.0130
0.33	0.39	0.0080	0.78	0.01	1.14	0.0120
0.67	0.43	0.0070	0.84	0.01	1.02	0.0920
1.00	0.35	0.0090	0.67	0.01	1.10	0.0130
1.50	0.30	0.0080	0.58	0.01	1.46	0.0130
2.00	0.23	0.0060	0.40	0.01	1.15	0.0100
4.00	0.14	0.0050	0.28	0.01	0.95	0.0100
8.00	0.10	0.0040	0.16	0.01	0.58	0.0090
12.00	0.06	0.0030	0.11	0.01	0.46	0.0120

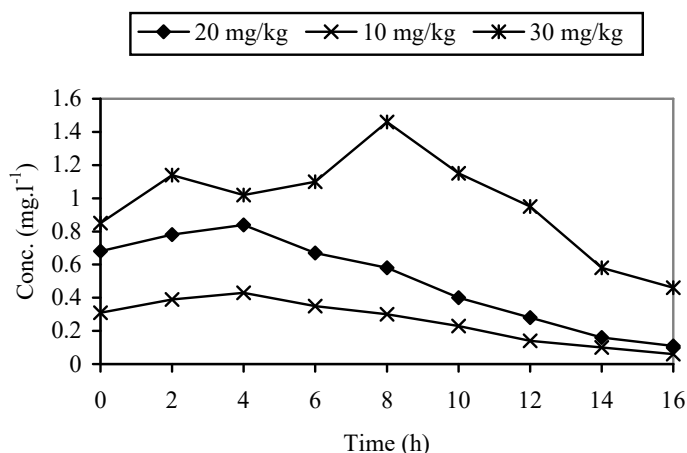


Fig 1: Time vs Plasma Level [Mean (n=12) $\pm$ SEM] profiles of ciprofloxacin in rabbits following oral administration of a single dose of 10, 20 and 30 mg/kg of body weight

Table 2
Pharmacokinetic parameter [Mean $\pm$ SEM (n=12)] of ciprofloxacin in rabbits following a single oral dose of 10, 20 and 30 mg/Kg body weight along with significance at (P<0.05)

PK Parameters	Unit	10 mg/kg	20 mg/kg	30 mg/kg	ANOVA*
Absorption Rate Constant	[h ⁻¹]	4.61 $\pm$ 0.61	4.23 $\pm$ 0.24	1.38 $\pm$ 0.32	Ns
Absorption Half-Life	[h]	0.17 $\pm$ 0.02	0.17 $\pm$ 0.01	0.71 $\pm$ 0.09	ns
Distribution Rate Constant	[h ⁻¹]	1.29 $\pm$ 0.11	1.04 $\pm$ 0.07	0.77 $\pm$ 0.29	ns
Distribution Half-Life	[h]	0.57 $\pm$ 0.04	0.7 $\pm$ 0.05	1.72 $\pm$ 0.23	ns
Elimination Rate Constant	[h ⁻¹]	0.12 $\pm$ 0.00	0.11 $\pm$ 0.01	0.07 $\pm$ 0.00	ns
Elimination Half-Life	[h]	5.73 $\pm$ 0.18	6.84 $\pm$ 0.87	10.94 $\pm$ 0.56	ns
AUC _(0-t) (Obs. Area)	[h.mg.l ⁻¹]	2.24 $\pm$ 0.07	4.47 $\pm$ 0.23	16.7 $\pm$ 0.53	**
AUC(0-inf.) (area)	[h.mg.l ⁻¹]	1.78 $\pm$ 0.04	3.31 $\pm$ 0.06	9.41 $\pm$ 0.08	**
MRT (area)	[h]	4.07 $\pm$ 0.06	3.98 $\pm$ 0.07	4.86 $\pm$ 0.03	**
C _(max) (calc)	[mg.l ⁻¹]	0.19 $\pm$ 0.01	0.34 $\pm$ 0.03	0.51 $\pm$ 0.03	ns
T(max)(calculated)	[h]	0.89 $\pm$ 0.07	0.91 $\pm$ 0.04	2.93 $\pm$ 0.32	ns
Lag Time	[h]	0.31 $\pm$ 0.06	0.21 $\pm$ 0.06	0.08 $\pm$ 0.16	ns

NULL Hypothesis that treatments (different dose levels) are equal was tested by ANOVA. Those found significantly different (P<0.05) are denoted by ‘***’ and those found non-significantly different are denoted by ‘ns’.

STATISTICAL ANALYSIS

All values are expressed as the mean $\pm$ standard deviation (SD) of twelve animals. The pharmacokinetics parameters obtained after the single oral dose of 10, 20 and 30 mg/Kg body weight were compared using a two way analysis of variance, considering a probability of P<0.05 to be significant.

RESULTS AND DISCUSSION

Plasma Levels:

Mean ciprofloxacin plasma concentration vs time at three different oral doses (10, 20, and 30mg/kg body weight) is given in Table-1 and presented in Fig. 1. Result obtained showed that plasma concentration of ciprofloxacin administered orally to

rabbits were best fitted to two compartmental open model, which is in agreement with the result obtained before in humans (Wise *et al.*, 1984) and in other species (Nouws *et al.*, 1988; Abadia *et al.*, 1994).

The maximum concentration ($C_{\max}$) obtained was 0.43 ± 0.007 , 0.84 ± 0.01 and 1.46 ± 0.93 ug/ml and were attained at 0.67, 0.67 and 1.50 for 10, 20 and 30 mg doses respectively.

When the dose administered increased, resulted in increased plasma concentration of ciprofloxacin, which is in agreement with the study of Boothe *et al* (2002), and $t_{\max}$ value were longer, indicating that absorption rate was slower and showed longer and more variable lag time, this has been already reported in humans (Gonzalez *et al.*, 1984) and in dogs (Walker *et al.*, 1990).

Pharmacokinetic Parameter

Pharmacokinetic parameter obtained after fitting the data to the statistical moment as shown in Table-2. Absorption rate constant, absorption half life, distribution rate constant, distribution half life elimination rate constant, and elimination half life did not show significant difference ($P < 0.05$), suggesting linearity of the kinetics.

On the other hand, MRT, $AUC_{(0-t)}$, and $AUC_{(0-\infty)}$ showed significant differences between the 30mg/kg and the other doses ($P < 0.05$), this suggesting the non linearity of ciprofloxacin pharmacokinetic following oral administration at doses higher than 20mg/kg body weight. As discussed by Abadia *et al* (1994) renal excretion of ciprofloxacin seems not to be impaired and disposition of ciprofloxacin after i/v administration was linear in dogs, and had no significant difference, would be related to absorption process or with non renal clearance of the ciprofloxacin (Abadia *et al.*, 1995).

In conclusion when ciprofloxacin is administered orally to rabbits at 10, 20, and 30 mg/kg body weight AUC dose data suggest significant difference between the doses which could be related with absorption

processes or with non renal clearance of the drug.

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